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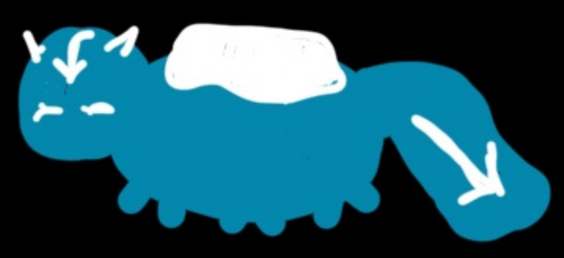
Chapter 5

According to Dr. Smith



Divisibility:

If $a, b \in \mathbb{Z}$ then we say a divides b & we write $a \mid b$ if $\exists\, c \in \mathbb{Z}$ st. $a \cdot b = c$.

$$a \mid b \neq \frac{a}{b}$$ → a & c are factors of b

we see $\underline{5 \mid 10}$ since $5 \cdot 2 = 10$.
10 is divisible by 5.

Why?

$\mathbb{N}, +, \cdot$

↳ building block is 1 or 0

↳ primes

$0, 1, 2, 3, 4 = 2 \cdot 2, 5, 6 = 2 \cdot 3, 7, 8 = 2^3$

It looks like numbers break into primes.

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Quotient Remainder Thm.

If $a, b \in \mathbb{Z}$ with $a \neq 0$, we write

$b = a \cdot q + r$ for a unique $q, r \in \mathbb{Z}$ with $0 \leq r < |a|$.

quotient remainder

ex

1. $a = 5$ $b = 10$ $\quad 10 = 5 \cdot \overset{q}{2} + \overset{r}{0}$

2. $a = 13$ $b = 141$: $\quad 141 = 13 \cdot 10 + 11$

$0 \leq 11 \leq 13$

pf

($a, b > 0$ if $b = 0$ then $0 = a \cdot 0 + 0$)

If $b \leq a$ then $b = a \cdot 0 + b$. This is nice b/c $0 \leq b < a$.

If $b = a$, then $b = a \cdot 1 + 0$.

Suppose $b > a$. Consider $S = \{b - xa : x \in \mathbb{N} \ \& \ b - xa \geq 0\}$

Note $S \subseteq \mathbb{N}$ and $S \neq \emptyset$ since $b - a \in S$.

By the WOP (well ordering principle) S has a least element $\to r_0$.

Note $r_0 = b \cdot x_0 a$ for some $x_0 \in \mathbb{N}$.

1st part of pf

We claim $0 \leq r_0 < |a|$.

For a contradiction ~~suppose $r_0 \geq a$~~, thus

2nd

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Note $r_0 = b - x_0 a$ for some $x_0 \in \mathbb{N}$.

1st part of pf

We claim $0 \leq r_0 < |a|$.

For a contradiction, suppose $r_0 > a$, thus

2nd (BWOC)

$r_0 - a \geq 0$.

We compute:

$$r_0 - a = b - x_0 a - a = b - (x_0 + 1) a$$

But $b - (x_0+1)a \in S$, & this contradicts r_0 being the least element.

Therefore $0 \leq r_0 < a$ as desired. We conclude since $r_0 = b - x_0 a$, that $b = x_0 a + r_0$ with $0 \leq r_0 < a$. $\square$

<u>Sketch of uniqueness:</u>

$b = aq_1 + r_1 = aq_2 + r_2$ with $0 \leq r_1, r_2 < |a|$

Then $aq_1 - aq_2 = r_2 - r_1$, so $a(q_1 - q_2) = r_2 - r_1$

$a \cdot c = b$

Since r_2 & r_1 are $< a$ & $\geq a$, we see w/some work $r_2 - r_1 = 0$ so $r_2 = r_1$.

Euclid's Idea:

Do quotient-remainder over & over again. We can find the greatest common divisor of a & b.

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Euclid's Idea:

Do quotient-remainder over & over again. We can find the greatest common divisor of a & b.

Euclidian Algorithm:

Given $a, b \in \mathbb{Z}$, we will find the gcd of a & b.

How?

$b = a \cdot q + r \quad 0 \leq r < |a|$

$a = r q_1 + r_1 \quad 0 \leq r_1 < r$

$r = r_1 \cdot q_1 + r_2 \quad 0 \leq r_2 < r_1$

$r_1 = r_2 \cdot q_2 + r_3 \quad 0 \leq r_3 < r_2$

$\vdots$

$r_3 = r_4 q_5 + r_5 \quad 0 \leq r_5 \leq r_4$

ex

gcd (738, 134)

$738 = 134 \cdot 5 + 118$

$134 = 118 \cdot 1 + 16$

$118 = 16 \cdot 7 + 6$

$16 = 6 \cdot 2 + 4$

$6 = 4 \cdot 1 + 2$ GCD

$4 = 2 \cdot 2 + 0$ The last non-zero remainder.

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Bézout's Theorem

If $a, b \in \mathbb{Z}$, then $\exists\ x, y \in \mathbb{Z} : ax + by = \gcd(a, b)$

$\gcd(788, 134)$

$788 = 134 \cdot 5 + 118$

$134 = 118 \cdot 1 + 16$

$118 = 16 \cdot 7 + 6$

$16 = 6 \cdot 2 + 4$

$6 = 4 \cdot 1 + 2$

$4 = 2 \cdot 2 + 0$

$118 = 788 - 134 \cdot 5$

$16 = 134 - 118 \cdot 1$

$6 = 118 - 16 \cdot 7$

$4 = 16 - 6 \cdot 2$

$2 = 6 - 4 \cdot 1$

*work back words

subbing in for remainders

$2 = 6 - (16 - 6 \cdot 2) \cdot 1$

$2 = \overset{1 \cdot 6}{6} - 16 + \overset{2 \cdot 6}{2 \cdot 6} \cdot 1$

$2 = \overset{3 \cdot 6}{3 \cdot 6} - 16$

$2 = 3(118 - 16 \cdot 7) - 16$

$2 = 3 \cdot 118 - 21 \cdot 16 - 16 = 3 \cdot 118 - 22 \cdot \underline{16}$

$2 = 3 \cdot 118 - 22(134 - 118 \cdot 1)$

$= 25 \cdot 118 - 22 \cdot 134$

$2 = 25 \cdot (788 - 134 \cdot 5) - 22 \cdot 134$

$= 25 \cdot 788 - 147 \cdot 134$